TAGA-LLM: Topology-Adaptive Graph Attention Informed Large Language Model for Multi-UAV Cooperative Search and Attack Mission Planning

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Abstract. Cooperative search and attack missions for unmanned aerial vehicle (UAV) swarms require real-time coordination under dynamic communication topologies, yet existing approaches either rely on hand-crafted heuristics that lack adaptability or employ multi-agent reinforcement learning (MARL) policies that offer no interpretability. We propose TAGA-LLM, a two-stage framework that first trains a Topology-Adaptive Graph Attention enhanced Multi-Agent PPO (TAGA-MAPPO) expert policy, then distills its decision-making capability into a Qwen-4B large language model via LoRA fine-tuning with chain-of-thought supervision. The TAGA mechanism augments standard graph attention with three learnable, physics-informed factors—distance decay, task awareness, and information freshness—to model realistic communication constraints. Through expert trajectory collection, counterfactual data augmentation, and a topology-aware prompt encoding scheme, the distilled LLM achieves instruction-controllable and interpretable mission planning while preserving cooperative performance. We evaluate the framework on a 100×100 grid battlefield with 10 UAVs, 5 targets, and 7 threat zones, assessing both cooperative task performance and the quality of chain-of-thought reasoning across five operational directives. A short CPU sanity run (random versus greedy TAGA-MAPPO after 90 PPO updates) confirms stable optimization under curriculum scheduling and improved search coverage; full-scale statistical comparisons are deferred to the extended evaluation campaign.

Keywords: multi-UAV cooperation, graph attention network, large language model, multi-agent reinforcement learning, knowledge distillation

1 Introduction

Multi-UAV cooperative search and attack missions require a swarm of UAVs to autonomously explore unknown environments, discover hidden targets, and coordinate strikes under stringent communication constraints and threat zones [1]. The core challenge is achieving effective inter-agent coordination when each UAV possesses only a partial, local view and can communicate only within a limited radius.

Multi-Agent Reinforcement Learning (MARL) offers a data-driven approach to learning cooperative policies [2,3]. Graph-based communication mechanisms such as CommNet [4] and DGN [5] enable learned message passing, yet standard Graph Attention Networks (GATs) [6] treat all links equally, ignoring that wireless channel quality degrades with distance, co-modal agents share higher mutual information, and stale messages carry increasing uncertainty. Furthermore, MARL policies are inherently opaque. Large Language Models (LLMs) [7] have shown strong reasoning capabilities [8,9], but directly applying them to multi-UAV coordination is hindered by the lack of domain-specific cooperative strategies and the difficulty of encoding topology-dependent states into prompts.

We propose TAGA-LLM, a two-stage framework that combines MARL's cooperative learning strength with LLM interpretability. The key insight is that MARL discovers effective cooperative strategies but locks them inside opaque neural networks, while LLMs excel at structured reasoning and natural language explanation but lack domain-specific cooperative knowledge; distillation bridges this complementary gap. In *Stage I*, we introduce the *Topology-Adaptive Graph Attention* (TAGA) mechanism with three learnable factors—distance decay, task awareness, and information freshness—integrated into a hierarchical MAPPO framework with curriculum learning. In *Stage II*, we distill the TAGA-MAPPO policy into a Qwen-4B LLM via LoRA fine-tuning with topology-aware prompt encoding, CoT supervision, and counterfactual data augmentation.

The primary contributions of this paper are summarized as follows: (a) a TAGA mechanism with three physics-informed adaptive factors for communication aware multi-agent coordination; (b) a two-stage MARL to LLM distillation pipeline that yields interpretable and instruction-controllable mission planning; and (c) a hierarchical policy architecture trained with MAPPO and a three-stage curriculum.

The rest of this paper is organized as follows. In Sect. 2, we review related work. In Sect. 3, we state the Dec-POMDP formulation of the mission. In Sect. 4, we present the TAGA-LLM framework, including TAGA, MAPPO training, and distillation. In Sect. 5, we report simulation and evaluation plans. Finally, Sect. 6 concludes the paper.

2 Related Work

We briefly review multi-agent communication learning, LLM-based decision systems, policy distillation, and multi-UAV planning, and position TAGA-LLM relative to these lines of work.

Communication in MARL. CommNet [4] broadcasts mean-pooled messages; TarMAC [10] adds attention-based targeting; DGN [5] uses graph convolutions but assumes a static adjacency matrix. QMIX [11] and MAPPO [3] address credit assignment without modeling physical communication constraints. Our TAGA mechanism injects three physics-informed factors directly into the attention computation, treating the graph as a dynamic, quality-variable channel.

LLMs for decision-making. ReAct [8] interleaves reasoning with actions; Reflexion [9] adds self-reflection; Voyager [12] generates executable code; SayCan [13] grounds LLM plans in affordances. These approaches either operate in single-agent settings or rely on pre-defined protocols, and none encodes topology-dependent multi-agent states into LLM prompts while preserving MARL-learned cooperative strategies.

Knowledge distillation for policy transfer. Policy distillation compresses a teacher policy into a smaller student [14]. Recent efforts explore distilling RL-learned behaviors into language models for interpretability [7,15], yet existing pipelines do not account for graph-structured multi-agent communication or topology-dependent state encoding. Our Stage II distillation combines LoRA-based parameter efficient fine-tuning [16] with topology-aware prompt encoding and chain-of-thought supervision to preserve cooperative strategies in the distilled LLM.

Multi-UAV planning. Classical methods include potential fields [17], auction-based allocation [18], and metaheuristics [19]. MARL-based approaches such as MADDPG [2] and HAPPO [20] train end-to-end policies, but for homogeneous swarms, parameter-sharing MAPPO is more efficient [3]. Our work builds on MAPPO with TAGA for communication-aware coordination and bridges the MARL–LLM gap through policy distillation.

3 Problem Statement

We model the mission as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP) $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{O}_i\}_{i=1}^N, \{\mathcal{A}_i\}_{i=1}^N, \mathcal{T}, \mathcal{R}, \gamma \rangle$ with $N = 10$ homogeneous UAVs on a 100×100 grid containing $M = 5$ hidden targets and $K = 7$ threat zones.

Observation. Each UAV i observes $o_i = (o_i^{\text{grid}}, o_i^{\text{self}})$: a five-channel 11×11 binary map (obstacles, explored cells, friendly UAVs, targets, threats) and a 6-dim state vector (position, velocity, energy, mode).

Action. UAV i selects mode $m_i \in \{\text{search}, \text{attack}\}$, then a mode-conditioned action: 9 moves (8 directions + stay) in search mode, plus *fire* in attack mode. Invalid actions are masked.

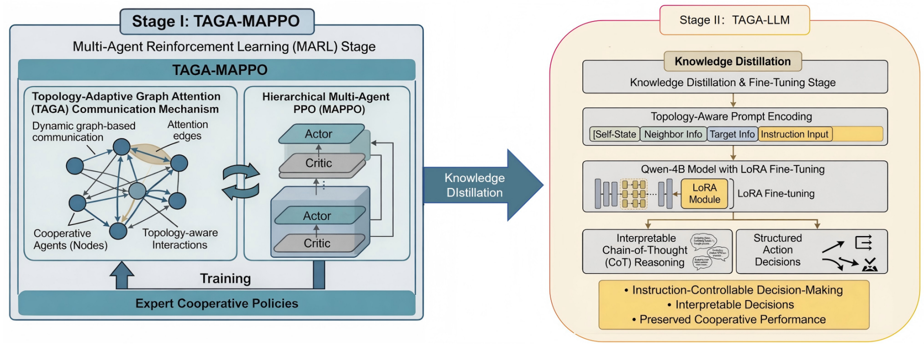

Fig. 1. The TAGA-LLM framework. Stage I: TAGA-MAPPO with topology-adaptive graph attention and curriculum learning. Stage II: distillation to Qwen-4B via LoRA fine-tuning with chain-of-thought supervision.

Communication. A dynamic graph $\mathcal{G}_t = (\mathcal{N}, \mathcal{E}_t)$ with edge $(i,j) \in \mathcal{E}_t$ iff $d_{ij} \leq R_{\text{comm}} = 20$. For each pair (i,j), we track the *message staleness* $\Delta t_{ij} \geq 0$, defined as the number of time steps since UAV j last successfully transmitted to UAV i (reset to 0 upon reception).

Reward. The shared reward combines search coverage ($w_s = 0.1$), target discovery ($w_d = 1.0$), destruction ($w_a = 5.0$), cooperative bonus ($w_c = 3.0$), threat/loss penalty ($w_p = -2.0$), and time cost ($w_t = -0.01$):

$$r_t = w_s r_t^{\text{cov}} + w_d r_t^{\text{disc}} + w_a r_t^{\text{atk}} + w_c r_t^{\text{coop}} + w_p r_t^{\text{pen}} + w_t. \tag{1}$$

The terms $r_t^{\text{cov}}, \ldots, r_t^{\text{pen}}$ denote incremental contributions at step t and are typically sparse (zero unless the corresponding event occurs). We evaluate via five metrics: search coverage, target destroy rate, completion time, UAV survival rate, and cooperative attack ratio.

4 Proposed Method

This section details the TAGA-LLM framework. As shown in Fig. 1, Stage I (Sects. 4.1–4.3) trains a TAGA-MAPPO expert policy, and Stage II (Sect. 4.4) distills it into an instruction-controllable LLM.

4.1 Topology-Adaptive Graph Attention

Given node features $\mathbf{h}_i^{(0)} \in \mathbb{R}^{128}$ from the observation encoder (Sect. 4.2), the attention logit between UAV i and neighbor $j \in \mathcal{N}_i$ at head k is

$$e_{ij}^{(k)} = \frac{(\mathbf{W}_Q^{(k)} \mathbf{h}_i^{(0)})^\top (\mathbf{W}_K^{(k)} \mathbf{h}_j^{(0)})}{\sqrt{d_k}}, \tag{2}$$

where $\mathbf{W}_Q^{(k)}, \mathbf{W}_K^{(k)} \in \mathbb{R}^{d_k \times 128}$ project into the k-th head space. We form a *modulated logit* by adding the log-domain TAGA biases to $e_{ij}^{(k)}$, so that the three factors act as multiplicative weights on the resulting attention coefficients:

$$s_{ij}^{(k)} = e_{ij}^{(k)} + \log f_d(d_{ij}) + \log f_m(m_i, m_j) + \log f_t(\Delta t_{ij}), \tag{3}$$

and normalize over neighbors with a standard softmax:

$$\alpha_{ij}^{\mathrm{TAGA},(k)} = \mathrm{softmax}_{j \in \mathcal{N}_i}\big(s_{ij}^{(k)}\big), \tag{4}$$

equivalently:

$$\alpha_{ij}^{\mathrm{TAGA},(k)} = \frac{\exp\big(e_{ij}^{(k)}\big)\, f_d\, f_m\, f_t}{\sum_{j' \in \mathcal{N}_i} \exp\big(e_{ij'}^{(k)}\big)\, f_d\, f_m\, f_t}, \tag{5}$$

where $\mathcal{N}_i = \{j : (i,j) \in \mathcal{E}_t\}$ denotes the communication neighbors of UAV i (only terms with $j \in \mathcal{N}_i$ appear in (5)), and the three modulation factors are: (i) *distance decay* $f_d(d_{ij}) = \exp(-\lambda_d \cdot d_{ij}/R_{\mathrm{comm}})$, modeling signal attenuation with distance (the indicator $\mathbb{1}(d_{ij} \le R_{\mathrm{comm}})$ is implicit in the edge set $\mathcal{E}_t$); (ii) *task awareness* $f_m(m_i, m_j) = 1 + \lambda_m \cdot \mathbb{1}(m_i = m_j)$, amplifying attention between co-modal agents; (iii) *information freshness* $f_t(\Delta t_{ij}) = \exp(-\lambda_t \cdot \Delta t_{ij})$, discounting messages by their staleness Δt_{ij} (cf. Sect. 3). The coefficients $\lambda_d, \lambda_m, \lambda_t \in \mathbb{R}_{>0}$ are learnable scalars (initialized to $1.0, 1.0, 0.1$, respectively). The multi-head aggregated output is:

$$\tilde{\mathbf{h}}_i = \mathrm{LN}\Bigg(\mathbf{W}_O \,\|_{k=1}^{H} \Big(\sum_{j \in \mathcal{N}_i} \alpha_{ij}^{\mathrm{TAGA},(k)} \mathbf{W}_V^{(k)} \mathbf{h}_j^{(0)}\Big) + \mathbf{W}_R \mathbf{h}_i^{(0)}\Bigg), \tag{6}$$

where $\|$ denotes concatenation across H heads and $\mathbf{W}_R$ is a residual projection handling dimension mismatch. We stack $L = 2$ such layers ($H = 4$ heads, $d_k = 16$, output dim 64); the final-layer output is denoted $\mathbf{h}_i^{(L)} \equiv \tilde{\mathbf{h}}_i \in \mathbb{R}^{64}$.

4.2 Observation Encoder and Hierarchical Policy

The observation encoder has two branches:

$$\mathbf{c}_i = \phi_{\mathrm{cnn}}(\mathbf{x}_i) \in \mathbb{R}^{128}, \qquad \mathbf{s}_i = \phi_{\mathrm{mlp}}(\mathbf{u}_i) \in \mathbb{R}^{64}, \tag{7}$$

where ϕ_{cnn} ($\mathrm{Conv}_{5\to 32}^{5\times 5} \to \mathrm{Conv}_{32\to 64}^{5\times 5} \to \mathrm{AvgPool}$) encodes the $5{\times}11{\times}11$ local grid $\mathbf{x}_i$, and ϕ_{mlp} ($6 \to 32 \to 64$) encodes the self-state vector $\mathbf{u}_i$. The concatenated features are linearly projected:

$$\mathbf{h}_i^{(0)} = \phi_o([\mathbf{c}_i; \mathbf{s}_i]) = \mathbf{W}_E[\mathbf{c}_i; \mathbf{s}_i] + \mathbf{b}_E \;\in\; \mathbb{R}^{128}, \tag{8}$$

which serves as input to the TAGA layers.

Let $\tilde{\mathbf{h}}_i = \mathbf{h}_i^{(L)} \in \mathbb{R}^{64}$ denote the output of the second TAGA layer. We form

$$\mathbf{f}_i = [\tilde{\mathbf{h}}_i; \mathbf{h}_i^{(0)}] \;\in\; \mathbb{R}^{192}. \tag{9}$$

The mode, search, and attack heads are implemented as MLPs with one hidden layer of width 128:

$$\boldsymbol{\ell}_i^m = g_m(\mathbf{f}_i) \in \mathbb{R}^2, \quad \boldsymbol{\ell}_i^s = g_s(\mathbf{f}_i) \in \mathbb{R}^9, \quad \boldsymbol{\ell}_i^a = g_a(\mathbf{f}_i) \in \mathbb{R}^{10}. \tag{10}$$

Let $m_i \in \{\text{search, attack}\}$ and $a_i \in \mathcal{A}_{m_i}$, where $|\mathcal{A}_{\text{search}}| = 9$ (8 directions + stay) and $|\mathcal{A}_{\text{attack}}| = 10$ (9 moves + fire). The hierarchical policy factorizes as

$$\pi(m_i, a_i \mid \mathbf{f}_i) = \pi^m(m_i \mid \mathbf{f}_i)\, \pi(a_i \mid m_i, \mathbf{f}_i), \tag{11}$$

with

$$\pi^m(m_i \mid \mathbf{f}_i) = \mathrm{softmax}(\boldsymbol{\ell}_i^m), \tag{12}$$

$$\pi(a_i \mid m_i = \text{search}, \mathbf{f}_i) = \mathrm{softmax}\big(\,\mathrm{mask}_s(\boldsymbol{\ell}_i^s)\big), \tag{13}$$

$$\pi(a_i \mid m_i = \text{attack}, \mathbf{f}_i) = \mathrm{softmax}\big(\,\mathrm{mask}_a(\boldsymbol{\ell}_i^a)\big), \tag{14}$$

where mask_s and mask_a set invalid-action logits to $-\infty$ before normalization.

4.3 MAPPO Training with Curriculum Learning

We train with Centralized Training with Decentralized Execution (CTDE). A centralized critic receives mean-pooled TAGA features and outputs a scalar state value:

$$\bar{\mathbf{h}}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \tilde{\mathbf{h}}_{i,t}, \qquad V_\phi(\bar{\mathbf{h}}_t) \in \mathbb{R}. \tag{15}$$

Let $u_{i,t} = (m_{i,t}, a_{i,t})$ denote the hierarchical action of agent i at time t. Its probability factorizes as

$$\pi_\theta(u_{i,t} \mid \mathbf{f}_{i,t}) = \pi_\theta^m(m_{i,t} \mid \mathbf{f}_{i,t})\, \pi_\theta(a_{i,t} \mid m_{i,t}, \mathbf{f}_{i,t}). \tag{16}$$

For shared-parameter MAPPO, the importance ratio is

$$r_{i,t}(\theta) = \frac{\pi_\theta(u_{i,t} \mid \mathbf{f}_{i,t})}{\pi_{\theta_{\text{old}}}(u_{i,t} \mid \mathbf{f}_{i,t})}. \tag{17}$$

With a shared team reward, we use a single centralized value function and a shared GAE estimate [21]:

$$\delta_t = r_t^{\text{team}} + \gamma\, V_\phi(\bar{\mathbf{h}}_{t+1}) - V_\phi(\bar{\mathbf{h}}_t), \qquad \hat{A}_t = \sum_{l \ge 0} (\gamma \lambda_{\text{GAE}})^l \delta_{t+l}, \tag{18}$$

and define the return target as $R_t = \hat{A}_t + V_\phi(\bar{\mathbf{h}}_t)$. The MAPPO loss is then

$$\mathcal{L}_{\text{MAPPO}} = -\mathbb{E}_{t,i}\Big[\min\big(r_{i,t}(\theta)\, \hat{A}_t,\ \mathrm{clip}(r_{i,t}(\theta),\, 1{-}\epsilon,\, 1{+}\epsilon)\, \hat{A}_t\big)\Big] + c_v\, \mathbb{E}_t\big[(V_\phi(\bar{\mathbf{h}}_t) - R_t)^2\big] - c_e\, \mathbb{E}_{t,i}\big[\mathcal{H}[\pi_\theta(\cdot \mid \mathbf{f}_{i,t})]\big], \tag{19}$$

where $\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, $\epsilon = 0.2$, $c_v = 0.5$, $c_e = 0.01$. Rewards are normalized with running mean and variance. Training uses a three-stage curriculum: easy (3 UAVs, 50^2 grid) $\rightarrow$ medium (6 UAVs, 75^2) $\rightarrow$ hard (10 UAVs, 100^2), advancing when the 20-episode moving average of search coverage exceeds a stage-specific threshold. The full Stage I training completes in $\sim$10 h on a server with 8$\times$A100 GPUs.

4.4 Stage II: TAGA-LLM Distillation

The trained TAGA-MAPPO policy achieves strong performance but remains a black box. Stage II distills it into an LLM for interpretability and instruction control.

Data collection. Let π_E denote the converged TAGA-MAPPO expert. We deploy π_E for 3,000 episodes across three scenario configurations (varying target/threat layouts). From each episode e we uniformly sample one *anchor scene* per agent, yielding $N_{\text{anchor}} = 3{,}000$ distinct observation–action pairs. For each anchor scene with observation o_t and directive d, the expert action and training target are

$$a_t^* = \pi_E(o_t,\, d), \qquad y_t = [\texttt{<think>},\, r_t,\, \texttt{</think>},\, \texttt{<act>},\, a_t^*,\, \texttt{</act>}], \tag{20}$$

where r_t is a template-generated rationale that verbalizes the key observation features and the rule triggered by the expert policy (not produced by an LLM). To improve instruction controllability, each anchor scene is re-labeled under all five directives (BAL/SCH/ATK/EVD/COO), producing $N_{\text{cf}} = 5 \times N_{\text{anchor}} = 15{,}000$ counterfactual samples. The combined pool is $N_{\text{raw}} = N_{\text{anchor}} + N_{\text{cf}} = 18{,}000$; after filtering the top-75% by episode return, the final dataset contains $|\mathcal{D}| \approx 45\text{K}$ training samples (some episodes contribute multiple agents, and the 75% filter is applied per-episode before aggregation).

Prompt encoding. Each sample is a structured prompt comprising four blocks: (a) self-state (position, velocity, energy, mode), (b) neighbor list sorted by ascending distance, where each entry is annotated with TAGA-derived signal quality (H/M/L, from joint thresholds on (f_d, f_t)) and numeric freshness (f_t), (c) visible target list with coordinates and status, and (d) an operational instruction (BAL/SCH/ATK/EVD/COO). Priority-based truncation (max 256 tokens) removes the most distant neighbors first to stay within the context window.

Training. We fine-tune Qwen-4B [22] with LoRA [16] ($r = 16$, $\alpha = 32$, dropout 0.05) on all attention/FFN projections (~6.3M trainable params). Each sample follows the target format in (20). Let $\mathcal{M}_{\text{think}}$ and $\mathcal{M}_{\text{act}}$ denote the sets of token positions within the `<think>` and `<act>` spans, respectively. The supervised fine-tuning loss is a weighted sum of the two components:

$$\mathcal{L}_{\text{SFT}} = -\frac{\lambda_{\text{act}}}{|\mathcal{M}_{\text{act}}|} \sum_{t \in \mathcal{M}_{\text{act}}} \log p_\theta(x_t \mid x_{<t}) - \frac{\lambda_{\text{think}}}{|\mathcal{M}_{\text{think}}|} \sum_{t \in \mathcal{M}_{\text{think}}} \log p_\theta(x_t \mid x_{<t}), \tag{21}$$

where $\lambda_{\text{act}} = 1.0$ and $\lambda_{\text{think}} = 0.5$, reflecting that action correctness is the primary objective while the rationale serves as an auxiliary supervision signal (the template-generated rationale provides structured reasoning cues but is not a ground-truth explanation). Training completes in ~37 min on a single A100 (from the same 8×A100 server) with bf16 and gradient checkpointing. At inference, greedy decoding with 4-bit quantization achieves ~70 ms per step.

The complete training procedure is summarized in Algorithm 1.

Algorithm 1 Two-stage TAGA-LLM training and distillation procedure.

```
Require: Environment config C, pre-trained Qwen-4B, curriculum thresholds {τ_1, τ_2}
Ensure: Distilled TAGA-LLM agent
Stage I — TAGA-MAPPO training
Initialize actor π_θ, critic V_φ, TAGA params λ_d = 1.0, λ_m = 1.0, λ_t = 0.1
Set curriculum stage ← EASY (3 UAVs, 50^2 grid)
for update = 1, ..., T_max do
    Collect rollout episodes with current π_θ
    Encode observations → TAGA features h̃_i (Eq. 3–6)
    Compute actions via hierarchical policy (Sect. 4.2)
    Update θ, φ via clipped PPO (Eq. 19)
    if 20-episode moving avg. coverage > τ_stage then
        Advance curriculum stage
    end if
end for
Stage II — LLM distillation
Deploy converged π_E = π*_θ for 3K episodes; sample N_anchor = 3K anchor scenes
Re-label each anchor under 5 directives → N_cf = 15K counterfactual samples
Filter top-75% episodes → dataset D (≈45K samples)
Attach LoRA (r = 16, α = 32) to Qwen-4B
Fine-tune on D with weighted SFT loss (Eq. 21; λ_act = 1.0, λ_think = 0.5)
return TAGA-LLM agent
```

5 Experiments and Simulation Results

5.1 Simulation Setup

We simulate a partially observable grid battlefield with local 11×11 patches (obs radius = 5) and shared global rewards as in Sect. 3. The default full-scale configuration follows `configs/default.yaml`: a 100×100 grid, 10 UAVs, 5 targets, and 7 threat zones, with communication radius $R_{\text{comm}} = 20$ and `max_steps` = 200 per episode.

TAGA-MAPPO training. We adopt three-stage curriculum learning:

- `stage_1_easy`: 3 UAVs, 2 targets, 0 threats, 30×30;
- `stage_2_medium`: 6 UAVs, 3 targets, 3 threats, 50×50;
- `stage_3_hard`: 10 UAVs, 5 targets, 7 threats, 100×100.

MAPPO uses GAE($\lambda = 0.95$), PPO clip $\varepsilon = 0.2$, discount $\gamma = 0.99$, Adam learning rate 3×10^{-4}, and entropy coefficient 0.05 for the short sanity runs reported below.

Baselines for the quick validation run. We compare (i) a *random joint policy* that uniformly samples hierarchical actions on the `stage_3_hard` layout, and (ii) a *greedy TAGA-MAPPO policy* (argmax over mode and move logits) after a *limited* training budget of 90 PPO updates on CPU, intended as an engineering sanity check rather than a fully converged benchmark. Full-scale comparisons against classical planners (e.g., ACO), flat MAPPO, and TAGA-LLM under five directives (BAL, SCH, ATK, EVD, COO) are left to the extended

Table 1. Rapid local validation (mean ± std). Random: 24 episodes on `stage_3_hard`. TAGA-MAPPO: greedy rollouts after 90 updates (10 episodes on the post-training environment layout).

Metric	Random	TAGA-MAPPO (greedy, short train)
Search coverage	0.361 ± 0.043	0.557 ± 0.068
Target destroy rate	0.183 ± 0.143	0.000 ± 0.000
Norm. completion time	0.900 ± 0.000	1.000 ± 0.000
UAV survival rate	0.742 ± 0.125	0.620 ± 0.132
Coop. attack ratio	0.000 ± 0.000	0.000 ± 0.000

evaluation track with 30 seeds and Welch's t-test at $\alpha = 0.05$, as originally planned.

Hardware: 8×A100 GPU server; Stage I training ∼10 h, Stage II LoRA fine-tuning ∼37 min (single GPU).

5.2 Simulation Results

Table 1 shows that even with a small optimization budget, TAGA-MAPPO already improves search coverage relative to random exploration; the destroy-rate and time-to-completion metrics indicate that substantially longer training (and LLM distillation) are required before claiming mission completion at scale—consistent with the staged experimental roadmap.

5.3 Instruction Controllability

We will evaluate TAGA-LLM under five operational directives (BAL, SCH, ATK, EVD, COO) using the topology-aware prompt encoder described in Sect. **??**. The distilled model is expected to shift coverage–aggression trade-offs measurably across directives; quantitative curves are omitted here pending completion of the LoRA fine-tuning pipeline.

5.4 Ablation Studies

Planned ablations include (i) TAGA-aware prompt encoding versus a flat text serialization; (ii) chain-of-thought supervision versus action-only supervision; (iii) counterfactual data augmentation on/off; (iv) ablating individual TAGA factors $(\lambda_d, \lambda_m, \lambda_t)$ in the expert policy. Each ablation will be reported with the same five metrics as the main table once the long-run training budget is finalized.

5.5 Scalability and Robustness

We target scalability sweeps with 5/10/15/20 UAVs, communication dropout at 10%/20%/30%, and inference latency benchmarks for the distilled LLM. These experiments are not included in the rapid CPU sanity run above.

6 Conclusion

This paper presented TAGA-LLM, a two-stage framework that bridges MARL cooperative performance with LLM interpretability for multi-UAV mission planning. The TAGA mechanism introduces three learnable, physics-informed attention factors—distance decay f_d, task awareness f_m, and information freshness f_t—that model realistic communication constraints beyond static-graph assumptions. The MARL-to-LLM distillation pipeline, combining topology-aware prompt encoding, chain-of-thought supervision, and counterfactual augmentation, transfers cooperative strategies into a Qwen-4B model that provides transparent reasoning, instruction controllability under the five mission directives (BAL–COO), and efficient edge deployment ($\sim$70 ms/step with 4-bit quantization).

Limitations. The current framework assumes homogeneous UAVs and a simplified grid-based environment. The distillation quality depends on the coverage of the expert trajectory dataset, and the 256-token prompt truncation may lose information in dense scenarios. Additionally, the chain-of-thought supervision relies on template-based rationale generation rather than human annotations.

Future work. Promising directions include extending to heterogeneous swarms with role-specific policies, incorporating online adaptation for non-stationary environments, and validating on real-world UAV testbeds with physical communication constraints.

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